

5 (a) Photosynthesis is a complex process involving a light dependent stage and a light independent stage.

(i) Name the products of the light dependent stage that are needed in the light independent stage.

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..... [2]

(ii) Describe the role of chlorophyll b in photosynthesis.

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A student carried out an experiment to investigate the effect of light intensity and light wavelength on the rate of photosynthesis.

- An aquatic plant, *Elodea canadensis*, was put into a beaker containing sodium hydrogencarbonate solution as a source of carbon dioxide.
- To minimise changes in temperature, an LED lamp was used as a source of light.
- The lamp was switched on and the number of bubbles released by the aquatic plant in 1 minute was counted.
- The lamp was placed at seven different distances from the beaker to change light intensity.
- Five replicates were carried out at each lamp distance.
- All other variables were controlled.

(b) The student calculated the light intensity for each distance ( $d$ ) using  $\frac{1}{d^2}$ .

Table 5.1 shows the calculated light intensities for each distance.

**Table 5.1**

| distance between plant and lamp / m | light intensity / $\frac{1}{d^2}$ |
|-------------------------------------|-----------------------------------|
| 0.025                               | 1600                              |
| 0.050                               | 400                               |
| 0.100                               |                                   |
| 0.150                               | 44                                |
| 0.200                               | 25                                |
| 0.250                               | 16                                |
| 0.300                               | 11                                |

Complete Table 5.1 by calculating the light intensity for distance 0.100 m.

[1]



- (ii) At a light intensity of 1600, explain why different colour filters result in different rates of photosynthesis.

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[Total: 11]